

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-inde  
xer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-  
a11107061ceea4298.jsonl`  
- SHA-256: `15f6e7411cf26983aa2dd24452f17f4e6bcbcb4dff758b4b72521b8b8d7d49142`  
- Source modified: `2026-06-06T17:31:46+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:31:11.948Z)

Source Extractor

Research question: "Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ? especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio)."

Fetch and extract key claims from this source:

****URL:**** <https://github.com/wywyWang/CoachAI-Projects/blob/main/LICENSE>

****Title:**** CoachAI-Projects LICENSE (hosts ShuttleSet) ? MIT License, (c) 2022 Wei-Yao Wang

****Found via:**** datasets/licensing search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-06T17:31:14.453Z)

I'll fetch the CoachAI-Projects LICENSE page to assess its content and licensing claims.

3. user (2026-06-06T17:31:31.182Z)

License Analysis

****License Type:**** MIT License

****Copyright Information:****

- Year: 2022
- Holder: Wei-Yao Wang

****Permission Grant Language:****

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4. assistant (2026-06-06T17:31:34.553Z)

The fetch succeeded and returned the full license content. This is the canonical LICENSE file from the CoachAI-Projects GitHub repository (the repo that hosts ShuttleSet), so it is a primary source for the licensing question.

5. user (2026-06-06T17:31:39.351Z)

Structured output provided successfully

6. assistant (2026-06-06T17:31:46.308Z)

Extracted the key licensing claims from the CoachAI-Projects LICENSE file. Summary:

- ****Source quality: primary**** ? this is the canonical LICENSE file in the GitHub repo that hosts ShuttleSet.
- ****Central claim:**** The repository is under the ****MIT License****, which permits commercial use, modification, and redistribution ? directly relevant to the project's "permissive / commercial-friendly license REQUIRED" constraint.

- **Copyright:** 2022, Wei-Yao Wang.

One caveat worth flagging for the parent research (not a claim, since the page doesn't state it): an MIT LICENSE file in the *repository root* governs the repo's *code*. Dataset assets like ShuttleSet (and any underlying broadcast video frames / derived annotations) can carry their own separate terms or third-party rights that the root MIT file does not automatically cover. The commercial-use conclusion is solid for the code; the dataset's redistribution/commercial status should be confirmed against ShuttleSet's own dataset card or paper before relying on it commercially.